Earth and Planetary Science Letters, 25 (1975) 177–188
© North-Holland Publishing Company, Amsterdam – Printed in The Netherlands

PALAEOMAGNETISM AND THE ROTATION OF ITALY

J.E.T. CHANNELL and D.H. TARLING

School of Physics, The University, Newcastle upon Tyne (Great Britain)

Received October 21, 1974
Revised version received December 11, 1974

New palaeomagnetic data from Cretaceous and Palaeocene rocks of peninsular Italy give pole positions which lie close to an inferred African polar wander curve and indicate that Africa and Italy have rotated as a single unit, during Mesozoic and Tertiary times.

1. Introduction

Over the last few years considerable interest has been centred around the possible movements of micro-continental blocks, such as Spain, Corsica, Sardinia and Italy, in the western Mediterranean [1–8]. The two main methods which have been used to reconstruct past positions of the major continents cannot be used very effectively in the Mediterranean. The method of least-squares fitting of continental shelf isobaths is not practicable in this region because distortions, caused by deposition, erosion and tectonic movements are significant when compared with the size of these fragments [9]. The use of sea-floor magnetic anomaly lineations [10] is not possible because they are either absent or obscured by the substantial thickness of post-Miocene sediments which often exceeds 4 km in the western Mediterranean. Thus, much of the evidence relating to the postulated rotations of these continental fragments has come from palaeomagnetism, however, the ages of tectonic settings of the rocks involved are often uncertain and Storetvedt [11] has suggested that the magnetic age of the rocks may often be considerably younger than the age of the rocks themselves. Until recently palaeomagnetic data from Italy have come only from the southern Alps and Sicily. The results of De Boer [12,13] and Van Hilten [14,15] from the Dolomites led to the postulation that the southern Alps, in Permian times, lay in the Himalayan region relative to Europe although this conflicted with palaeontological evidence for the proximity of southern Europe and north Africa. More recent palaeomagnetic data from Permian rocks in stable (extra-Alpine) Europe and the southern Alps [8] remove the evidence for this megatectonic movement. Nonetheless, Permian and Triassic results from the southern Alps [4,7] when compared with results from stable Europe differ by 50° in declination. This has been widely accepted to indicate a post-Late Triassic 50° anticlockwise rotation, relative to Europe, of an Italian "microplate". Soffel [6] has suggested that most or all of this rotation occurred between the Late Eocene and Late Miocene, based on palaeomagnetic work from Colli Euganei in the southern Dolomites. Schult [16] and Barberi et al. [17] sampled Upper Cretaceous basalts at Capo Passero in the relatively stable block of southeast Sicily. Their palaeomagnetic results are consistent with African data of the same age and this evidence, together with seismic crustal studies in the Sicilian – Tunisian sector of the Mediterranean [18], indicates that Ragusei – Iblei carbonate platform in southeast Sicily is a prolongation of the north African platform, and no significant post-Cretaceous relative movement has occurred between Africa and this part of Sicily.

All previous Italian palaeomagnetic results therefore came from the southern Alps or Sicily and this

study was designed to provide data from peninsular Italy. However, much of the Apennines is allochthonous so that individual tectonic units may have rotated relative to each other, masking any bulk rotation and there are virtually no red siltstones or igneous rocks of suitable age that would normally be the basis for a palaeomagnetic study. Instrumental improvements [19] now allow the use of more weakly magnetic rock types. Very recently Lowrie and Alvarez [20] have sampled the Scaglia deposits of the Northern Apennine Umbrian sequence and determined a 43° anticlockwise rotation of this part of the Italian "microplate" relative to stable Europe since the Late Cretaceous. The same sequence, together with other sequences in the southern Dolomites and southern Apennines, were samples in 1973 and 1974 and the results, presented here, indicate a different interpretation.

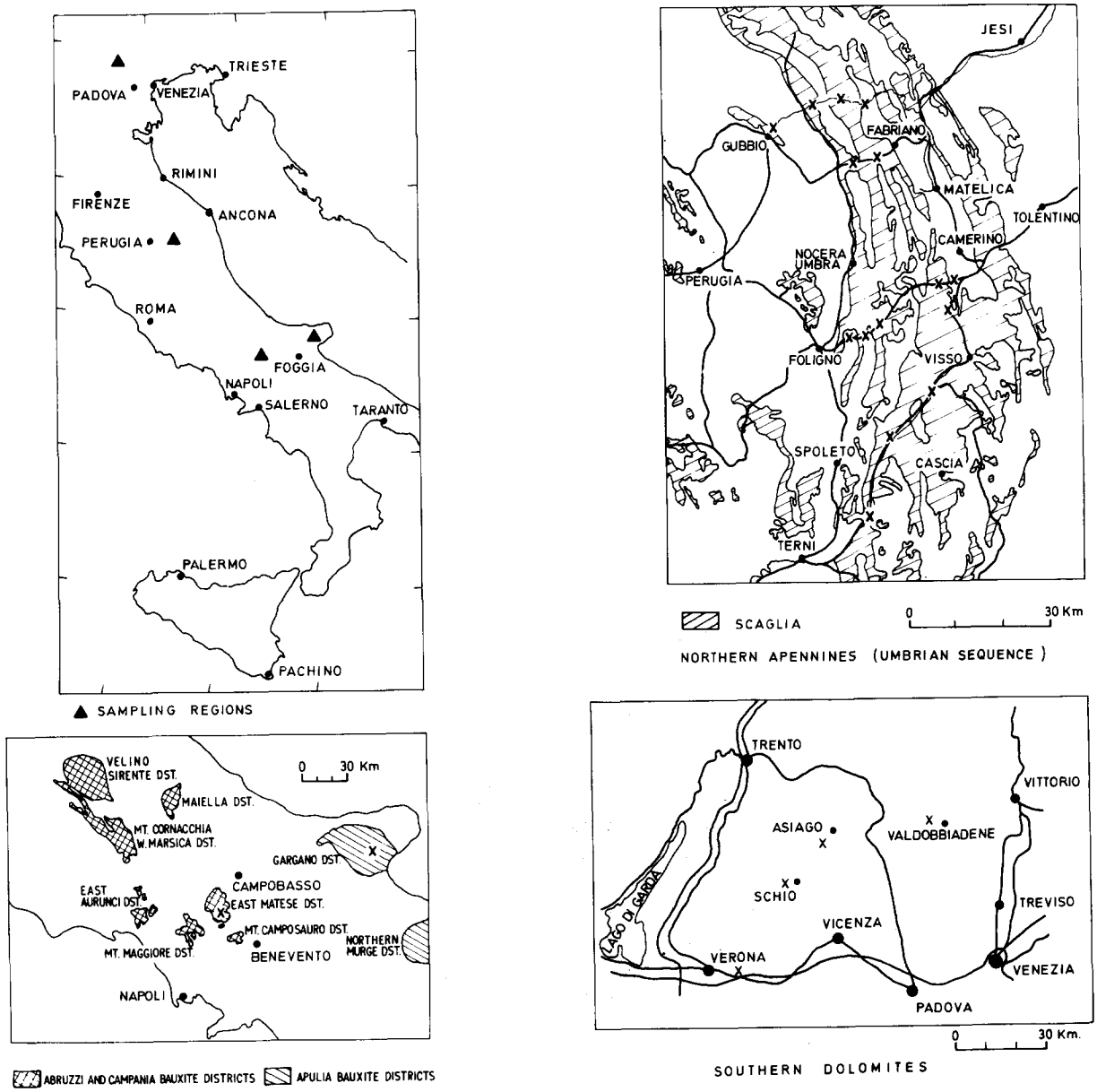


Fig. 1. Location of the sampling regions.

2. Northern Apennines

The Umbrian sequence of the Apennines (Fig. 1) is considered autochthonous or para-autochthonous [21] and also contains a highly magnetically stable horizon. The scaglia horizon of the Umbrian sequence has a total thickness of 210–470 m [22] and comprises fine-grained deep-water marly limestones which are red, white or, rarely, green in colour. The Scaglia outcrops sporadically along a zone 60 km wide which runs from Urbino in the north to Rieti in the south, a distance of some 200 km. Renz [23,24] distinguished four biozones within the Scaglia which range in age from Cenomanian to Early Eocene. The principal zone that was sampled was the Scaglia Rossa which was originally given an age of Turonian – Campanian [23,24], but is now considered to extend into the Palaeocene [22].

Forty-nine sites were collected over about 100 km (N–S) and 50 km (E–W) (Table 1 and Fig. 3). Thirty-three of the sites were from red Scaglia and the rest varied in colour from white/grey to green. The limestones had initial intensities of remanence between 10^{-6} and 10^{-7} gauss and all showed magnetic stability on thermal demagnetisation up to 600°C (Fig. 2). Six of the sites (7, 22, 24, 36, 37, 49) were rejected on the grounds of poor within-site grouping ($k < 10$, $\alpha_{95} > 20$). Two sites (20, 35) were eliminated due to their low magnetic intensity, and sites 30–34 and 42, 46–47 were rejected because they came from regions where local structural distortion did not allow sufficiently accurate tectonic corrections to be made. Sites 9–14 were from Lower Cretaceous (Aptian) white limestones beneath the Scaglia. All other sites were assumed to have an age within the range Senonian to Late Palaeocene. The means of the 27 Upper Cretaceous/Palaeocene sites were well grouped (k = 19.0, α_{95} = 6.5) but there was a marked spread in declination. Part of this spread could be due to tectonic rotations occurring during the age range of the samples, bu the mean of the magnetic directions should nevertheless give a good representation of the magnetic direction pertaining to the Latest Cretaceous and Early Palaeocene. Lowrie and Alvarez [20] have also sampled 22 sites from the Scaglia Rossa of the Umbrian sequence and obtained a direction of 320.1°, + 45.4° (α_{95} = 5.9).

3. Southern Dolomites

The area sampled is in the southern Dolomites region, in the Lessinian and Vicentinian Alps (Fig. 1). Although the region is heavily influenced by faulting, there are no indications of large-scale allochthonous movement. The main horizon sampled is similar in age and lithology to that sampled from the Umbrian sequence and comprises the Senonian Red Scaglia from which 17 sites were collected from three localities – Illasi (Verona), Schio and Valdobbiadene. A further four sites (4, 5, 20, 21) were collected from Cenomanian white limestones at Illasi and Valdobbiadene and two sites (6, 14) from Tertiary basalts. Two sites (17, 18) from the Jurassic "Ammonitico Rosso" formation near Asiago were found to be unstable.

Most of the sites exhibit magnetic stability on thermal demagnetisation (Fig. 2) and only three sites (23–25) were rejected due to poor stability. Site 10 was rejected due to its poor within-site grouping and site 20 was too weakly magnetic for precise measurement. Sites 6, 14 and 15 are basalts of uncertain age, probably Tertiary, and cannot be considered for the ascertation of the Cretaceous directions (Table 2, Fig. 3). The concordance of the results from these three regions bears out the assumption that little rotation has occurred between the different units and that the mean result can be used to ascertain the rotation of the southern Dolomites relative to Europe and Africa.

4. Apulian platform

Reconstructions of the stages of the late Tertiary crustal shortening in the southern Apennines [25] show that the Apulian platform can be considered autochthonous so that palaeomagnetic results from this region should give magnetic directions uninfluenced by local Tertiary deformations. Unfortunately, the Apulian platform consists almost entirely of Mesozoic shelf-carbonate deposits which cannot be used for paleomagnetic study due to their very weak remanence ($< 5 \times 10^{-8}$ gauss). However, there are two principal localities (Fig. 1) where Cretaceous bauxites are interbedded with the platform limestones. One such deposit is at San Giovanni Rotondo in the Gargano district to the northeast of Foggia.

TABLE 1
Umbrian sequence (Perugia) – after thermal cleaning

Site	N	Stability index	Intensity ($\times 10^{-6}$ gauss)	Dec.	Inc.	α_{95}	k	^{o}N	^{o}E
1	6	6.4	0.25	165.2	−32.2	6.4	110.2	−61.3	43.3
2	7		0.30	149.6	−34.6	6.3	93.62	−54.3	68.6
3	6	3.58	0.15	154.0	−18.6	13.9	24.08	−49.5	54.6
4	6		0.07	139.5	−12.1	25.5	7.87	−38.5	68.5
5	8		0.80	297.1	25.1	5.1	119.12	28.8	273.5
6	9	4.65	0.50	148.0	−40.6	3.9	176.06	−56.9	75.8
7*	6	3.00	0.25	273.1	22.7	19.9	12.24	11.3	288.9
8	9		0.20	314.9	17.9	11.3	21.83	38.3	255.4
9	6	2.32	0.20	304.6	23.3	6.2	119.10	33.3	266.9
10	6	3.57	1.40	300.5	43.0	6.2	116.99	38.6	282.4
11	6	3.65	2.50	291.0	35.2	2.0	1154.3	28.3	283.4
12	6		0.70	291.1	45.4	7.7	75.88	32.9	290.3
13	6	5.77	2.40	293.4	41.6	5.9	129.78	32.7	286.1
14	8		0.50	306.9	48.2	5.0	122.58	46.0	282.7
15	6		0.60	335.8	18.6	20.0	12.18	50.1	233.1
16	7		0.80	324.9	37.6	5.6	116.58	53.5	256.7
17	7	1.87	0.80	125.7	−18.4	7.6	64.6	−32.2	83.7
18	6		1.50	333.2	41.3	4.1	267.14	60.5	249.6
19	6		0.45	147.4	−30.1	6.9	94.00	−51.2	68.5
20*	6	1.46	0.07	115.6	20.4	14.5	22.42	−10.4	76.8
21	5		0.05	309.8	32.4	25.5	9.95	40.9	269.7
22*	7		0.45	296.1	12.0	26.4	8.58	23.2	268.8
23	7	7.82	0.70	331.0	34.5	5.8	110.23	55.6	247.0
24*	7		0.30	143.7	−48.9	20.1	9.94	−57.9	100.0
25	7		1.70	314.7	54.5	5.0	147.10	54.2	285.6
26	6	0.15	0.15	22.7	39.3	4.5	223.67	62.2	143.0
27	5	3.36	0.80	346.5	37.5	5.3	205.82	65.4	224.1
28	8		1.00	346.7	30.1	8.0	49.20	61.3	219.7
29	6	3.91	1.90	1.2	23.3	9.6	49.65	59.5	190.4
30*	6		0.40	207.8	32.8	6.1	120.4	−24.0	343.8
31*	6	10.01	0.50	17.7	49.0	6.0	124.04	71.0	139.5
32*	6	5.63	0.80	50.6	−22.8	9.8	48.03	+17.5	140.4
33*	7	2.87	0.40	335.1	−36.1	7.3	68.47	22.9	217.8
34*	8		0.25	2.4	39.2	10.6	28.13	69.8	186.0
35*	6		0.03						
36*	9		0.50	149.8	−17.0	90	1.50	−41.2	48.5
37*	8	2.38	0.20	184.1	−20.8	19.8	8.75	−58.1	3.9
38	8		0.60	336.8	39.6	5.0	125.86	62.0	243.5
39	4		0.40	328.2	40.9	8.7	111.85	57.4	256.6
40	9		0.70	325.3	32.3	2.2	541.55	51.2	252.9
41	6	2.13	0.25	302.6	22.1	13.4	26.12	31.7	268.5
42*	7		0.15	17.0	36.6	7.7	61.85	63.4	154.8
43	4		0.30	144.4	−48.1	10.0	85.89	−58.5	89.4
44	5	2.53	0.40	136.2	−42.2	6.0	162.78	−49.9	89.9
45		3.31	1.20	316.8	35.2	13.5	33.28	47.1	263.6
46*	6		0.20	0.3	30.9	11.9	32.57	64.1	192.7
47*	6		0.30	153.2	7.5	13.1	27.02	−37.4	47.7
48	10	5.28	0.50	355.7	40.7	4.5	114.42	70.3	204.7
49*	8	1.07	0.08	269.4	−21.2	34.0	3.62	− 9.1	270.3
Upper Cretaceous/Palaeocene									
Mean:	27			328.8	33.9	6.5	19.0	53.9	249.2
Middle Cretateous									
Mean:	6			297.9	39.6	8.7	59.5	35.2	281.4

* Eliminated sites.

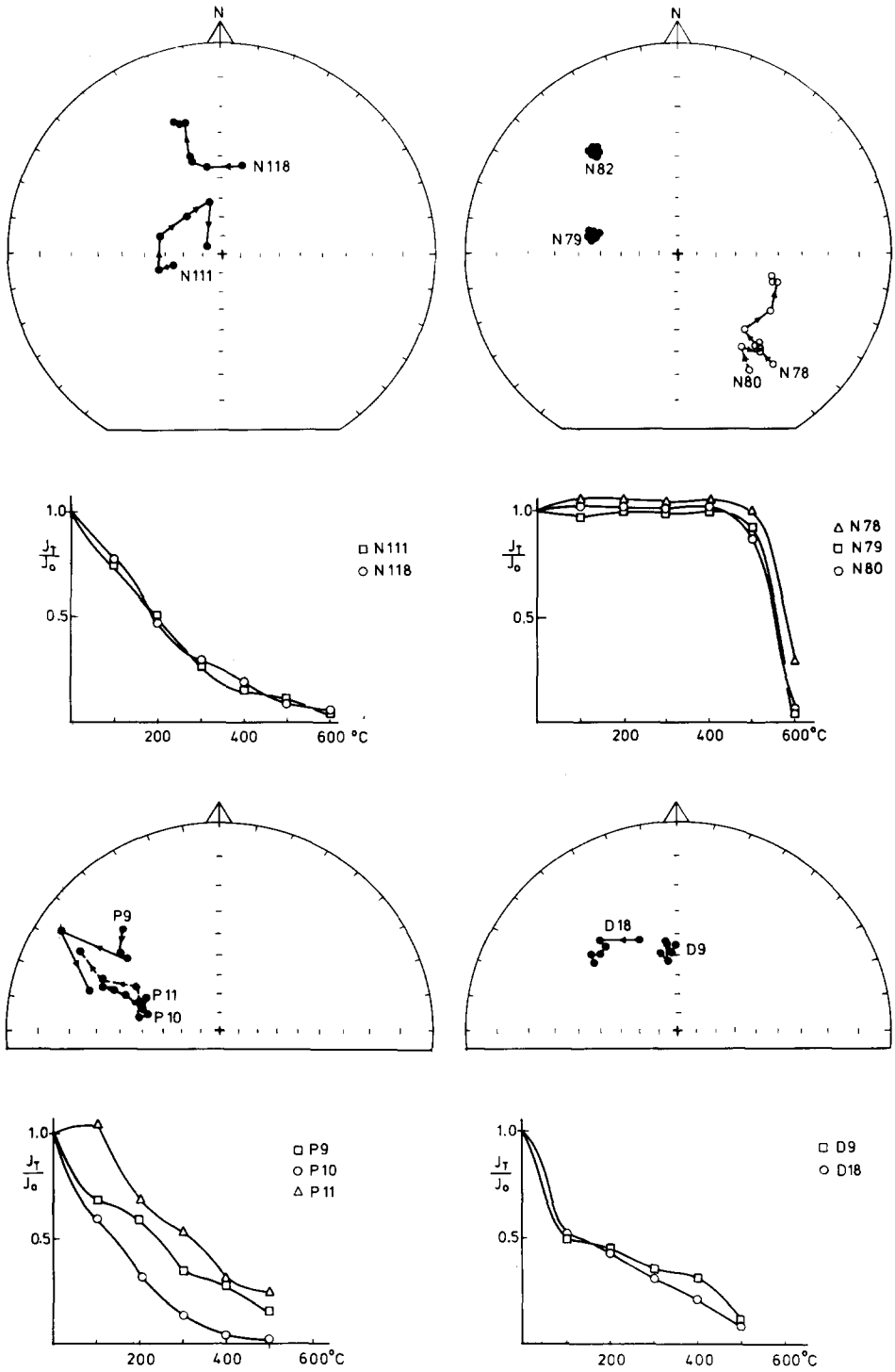


Fig. 2. Representative demagnetisation characteristics of the Matese Mountain bauxites (N78–80, 82) Apulian bauxites (N111, 118), Umbrian sequence Scaglia (P9–11), southern Dolomites Scaglia (D9, 19).

TABLE 2
Dolomites – after thermal cleaning

Site	N	Stability index	Intensity ($\times 10^{-6}$ Gauss)	Dec.	Inc.	α_{95}	k	o_N	o_E
1	8		7.50	333.6	48.1	3.8	212.11	63.7	252.4
2	10		6.50	344.0	58.2	4.4	119.67	77.0	258.5
3	10	6.36	6.50	9.0	42.8	3.1	277.59	68.2	168.9
4	6	3.31	0.40	334.3	33.7	7.3	86.32	55.6	238.2
5	5		0.20	332.8	46.1	14.3	29.43	62.5	250.4
6*	8	12.91	4000	168.6	−18.9	6.8	67.24	−52.9	30.3
7	11	4.90	1.30	330.0	43.9	4.8	90.76	59.1	252.4
8	8		1.00	341.3	35.8	5.8	92.69	60.3	228.3
9	6	4.87 6.21	0.90	345.3	44.9	4.6	209.87	67.5	228.0
10*	9	2.68	0.70	339.9	13.6	82.2	1.36	49.1	183.5
11	8		2.50	332.7	33.9	4.2	172.02	54.8	240.2
12	8	3.80	0.60	340.7	41.5	5.5	101.21	63.3	234.0
13	5		2.00	332.3	33.8	6.3	147.72	54.6	240.6
14*	5	7.56	100.0–2000.0	106.2	40.8	9.6	64.33	6.2	73.6
15*	7		4.0–400.0	80.4	34.7	1.0	3883.6	20.1	94.3
16	6		3.50	327.6	26.8	12.4	29.93	48.9	242.4
17*	6	0.90	0.-5	339.4	69.5	48.3	2.87	71.9	307.9
18*	7	6.01	0.60	263.4	8.4	90	1.31	0.4	288.1
19	9		11.00	350.5	42.1	5.5	88.36	67.3	215.6
20*			0.01						
21	7		0.15	145.3	−28.9	8.6	50.47	−48.1	67.3
22	8	4.47	1.40	13.5	35.1	9.4	35.38	61.8	164.0
23*	7	0.70	0.25	290.8	12.1	46.9	2.61	22.1	271.7
24* 25*	unstable								
Upper Cretaceous/Palaeocene									
Mean	12			343.0	41.4	7.5	33.8	64.1	229.2
Middle Cretaceous									
Mean	3			330.5	36.5	14.9	69.5	54.8	244.6

* eliminated sites

The disused bauxite mine at San Giovanni Rotondo is in an east–west line of bauxite pockets about $1\frac{1}{2}$ km long. The deposit has a maximum thickness of 8 m and rests on an eroded surface of Aptian–Albian calcilutite (26). The following sequence of events is visualised for the formation of the bauxites:

(1) Elevation of this part (the Gargano district) of the Apulian platform.

(2) Formation of the eroded limestone surface and soil development.

(3) Differential subsidence.

(4) Transport of soils from the more elevated regions into a shoreline shallow sea environment to fill the cavities in the eroded surface of the limestone.

Thin clayey layers at the same stratigraphic level as the bauxites are found in the Gargano hills and this probably represents the remains of the source material of the bauxites.

The bauxites vary in nature throughout their thickness. At the base there is usually a soft clayey layer, above this the bauxite is mainly pisolitic. Towards the top of the pockets the proportion of reworked bauxite fragments and the proportion of calcite increases. Immediately above the bauxites there is a sporadic

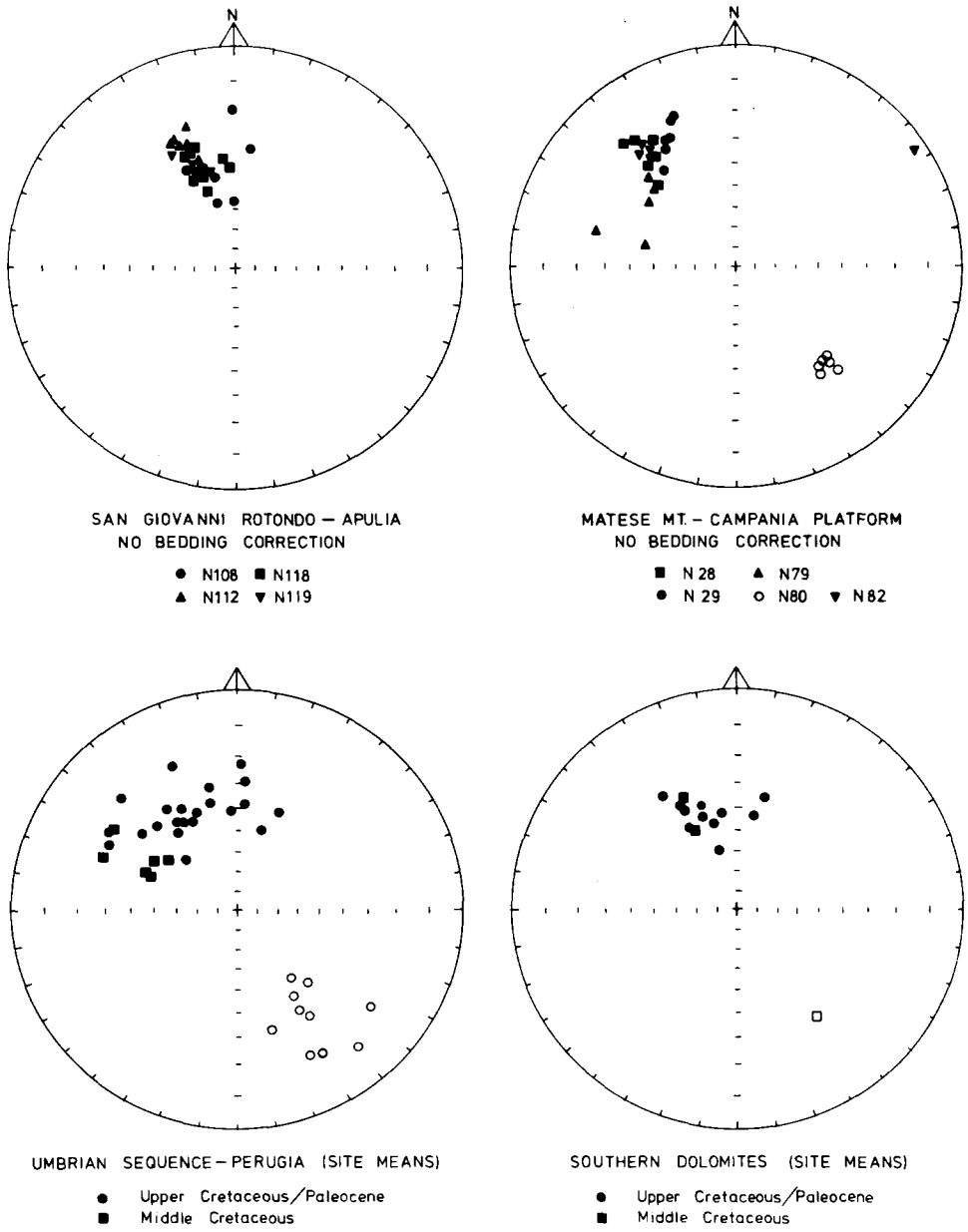


Fig. 3. Magnetic directions after thermal cleaning. Sample directions are plotted for Matese Mountain and Apulia, and site means for the southern Dolomites and the Umbrian sequence.

thin layer of dirty limestones containing bauxitic pigment. The mode of origin of the bauxites is important when considering the tectonic correction to apply to the palaeomagnetic directions. The bedding laminations in the bauxite are unlikely to give a good estimate of the original horizontal as much of the deposition has occurred on the inclined walls of the pockets. Although the Senonian calcilutites which overlie the bauxite are slightly disconformable [26], the underlying limestones appear to be close to the horizontal, so no tectonic correction was made to the bauxite results.

TABLE 3
Apulia – after thermal cleaning

Site	N	Stability index	Average intensity ($\times 10^{-6}$ gauss)	Dec.	Inc.	α_{95}	k
20	8	1.02	0.04			39.5	2.92
21	7	1.95	3.0			24.1	5.53
22	6	3.55	15.0	326.6	36.6	27.0	7.12
108*	6	5.28	5.0	352.6	40.7	14.4	22.60
109	6		0.05				
110	7	4.08	10–100			60.4	1.95
111	7	1.42	5.0	282.6	51.6	21.5	10.62
112*	6	10.09	0.60	337.1	27.2	4.4	231.4
113	7	4.63	20			49.4	2.45
114	5	6.16	100			54.8	2.91
115	9		0.07				
116	12	8.32	5000			36.7	2.36
117	12	2.78	10.0			30.3	3.02
118*	9	8.99	3.0	343.0	38.1	5.3	94.39
119*	5	1.84	1.0	337.4	36.8	6.2	151.1
120	6	23.99	3.0			90	1.42
121	5	6.36	0.2			55.0	2.89

Middle Cretaceous mean direction from the four usable sites (*) giving equal weight to each sample. $D = 342.4$. $I = 36.0$, $k = 46.2$, $\alpha_{95} = 4.2$; pole position: lat. = 63.7°N; long = 235.5°E.

Seventeen sites were taken from San Giovanni Rotondo, with six separately oriented samples drilled from each site (Table 3 and Fig. 3). Seven sites were in dirty limestones at the top of the bauxites and ten from the bauxites themselves. The clayey material at the base of the bauxites was too soft to sample efficiently and, towards the top, where calcite tends to bind the material, the proportion of reworked bauxite fragments is high. All sites were magnetically stable or very stable to thermal demagnetisation, with stability indices [27] varying from 1.50 to 25 during demagnetisation up to 600°C (Fig. 2), but only two of the bauxite sites and two of the limestone sites displayed good within-site grouping ($k > 15, \alpha_{95} < 15$). The bauxites which exhibit unacceptable grouping are those which include a high proportion of reworked bauxite intraclasts. The size of the individual fragments relative to that of the measured sample is such that the remanence of the fragments is unlikely to be nullified throughout the sample so that the remanence due to the intraclasts influences the magnetic direction associated with the matrix. This inhomogeneity varied from sample to sample and accounts for the poor within-site grouping. The only two useful bauxite sites were pisolitic and contained little or no reworked material. Three of the limestone sites had to be rejected because of their low magnetic intensity and the other two discarded limestone sites contained sufficient reworked bauxite material to produce magnetic inhomogeneity and low within-site grouping.

5. Campania platform

The other major bauxite deposit in Apulia is at Spinazzola in the Northern Murge district (Fig. 1). Three sites were collected from this deposit for initial investigation but the within site dispersion was unacceptably high, again probably due to the high intraclastic content of the samples.

Although the Abruzzi–Campania platform has undergone allochthonous movement, the fault pattern does not indicate any large-scale rotation relative to the Apulian platform (B.D'Argenio, personal communication) so that palaeomagnetic results from this re-

TABLE 4

Campia (Matese Mt.) – after thermal cleaning

Site	N	Stability index	Dec.	Inc.	Intensity ($\times 10^{-6}$ gauss)	α_{95}	k
28*	6	5.2	321.4	24.9	5	6.7	101.5
29*	6	16.8	332.0	24.6	100	5.9	131.4
30	8	0.7			0.6		
78	6	6.72	301.6	26.6	2.50	16.2	17.9
79*	5	22.1	300.7	35.7	90.0	14.1	30.47
80*	6	17.5	139.3	−29.1	10.2	2.8	590.7
81	7	4.07	264.1	2.5	10.0	27.1	7.08
82*	5	20.7	321.6	24.6	60.0	3.7	614.4
83	10	18.3	348.9	53.7	20.0	6.5	56.24
84	5	0.69			1.00		
85	6	0.53			0.07		
86	8	7.06	342.6	47.2	15.00	19.9	8.72
87	5	0.60	350.7	46.4	0.60	19.4	16.47

Middle Cretaceous mean direction from the five usable sites (*) giving equal weight to each sample. $D = 322.4$, $I = 28.2$, $k = 16.1$, $\alpha_{95} = 7.0$; pole position: lat = 48.2°N; long 256.7°E.

gion may be expected to be more or less concordant with the Apulian result and therefore applicable to the Apulian block as a whole.

Thirteen sites were sampled from the bauxites and limestones of Matese mountain, which is part of the Campania platform (Table 4 and Fig. 3). The bauxite is wedged between the same stratigraphic units as that at San Giovanni Rotondo and is assumed to be of the same age. The deposit has a maximum thickness of 2.5 m and was again principally pisolitic with reworked bauxite intraclasts. The nature of the material is more consistent throughout its thickness than that at San Giovanni Rotondo. Twelve of the sites were from bauxites, the other site (30) came from the overlying Senonian limestone, but this was too weakly magnetic for precise measurement. Three of the bauxite sites (84, 85, 87) were found to be magnetically unstable, and the stable component characteristic of the remaining nine sites is shown in Fig. 2. Two of the bauxite sites (83, 86) were remagnetised to the present earth's field, and one site (78) exhibited dual polarity and was therefore rejected on the grounds of probable remagnetisation. The bauxite from site 81 contains a marked concentration of reworked bauxitic material and this site was rejected due to its poor within-site grouping.

6. Interpretation

A major problem in any palaeomagnetic interpretation is that there are few reliable data from stable Europe or Africa of Cretaceous or Jurassic age against which direct comparisons can be made. Van der Voo and French [28] have used the Atlantic magnetic anomaly patterns [10], together with selected palaeomagnetic data, to derive a series of reconstructions of the Atlantic-bordering continents since the Late Carboniferous. This method provides putative pole positions for stable Europe and Africa for the Jurassic and Cretaceous that depend mainly on the validity of North American palaeomagnetic data and the accuracy of the magnetic anomaly fit. Nonetheless, until more palaeomagnetic data are obtained from Europe and Africa these putative pole positions are the most reliable available and allow an evaluation of the rotations of the sampled regions relative to these continents. The required rotation (Table 5) of the Dolomites and Apulia in order that the Cretaceous magnetic directions fit the African polar wander curve (Fig. 4) is very small as is that deduced for Sicily [16,17] and for Malawi [29]. Similarly, the Permian and Triassic observations from the southern Alps [4,7] agree well with the African polar wandering curve. The Umbrian

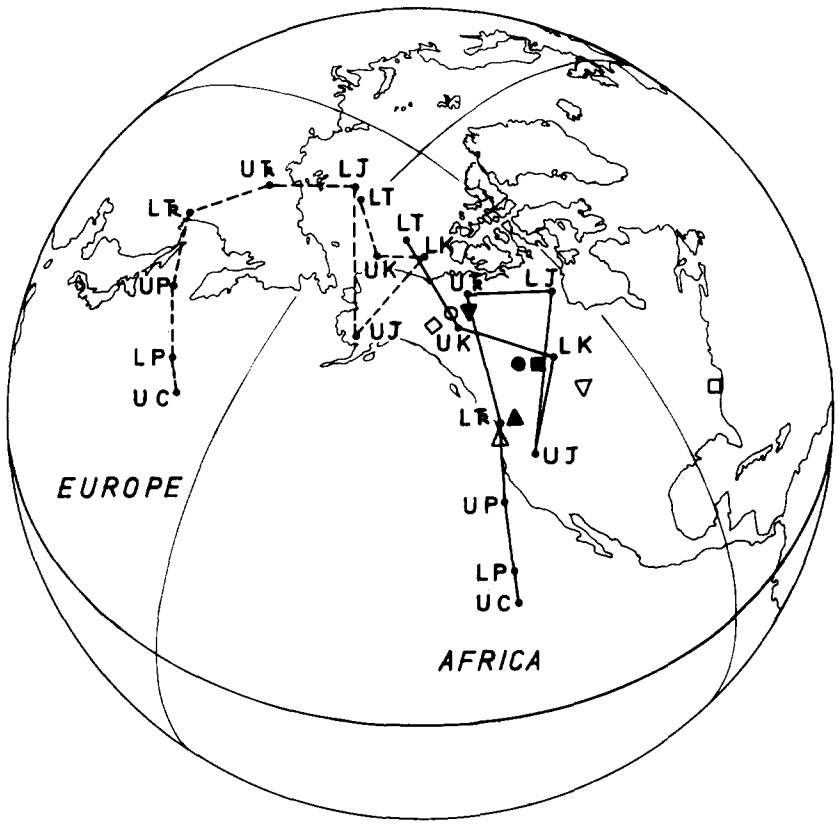


Fig. 4. Polar wander curves for Europe and Africa from Upper Cretaceous (U.C.) to Lower Tertiary (L.T.) from the data of Van der Voo and French [28]. The six new Italian pole positions are plotted together with those of other workers. Key: ▲ = Dolomites Upper Triassic [4]; △ = Dolomites Lower Permian [7]; ◇ = Sicily Upper Cretaceous [16,17]; ■ = northern Apennines Upper Cretaceous/Palaeocene; □ = northern Apennines Middle Cretaceous; ● = Dolomites Middle Cretaceous; ○ = Dolomites Upper Cretaceous/Palaeocene; ▼ = Apulia Middle Cretaceous; ▽ = Campania Middle Cretaceous.

sequence from the northern Apennines and the Campania platform of the southern Apennines, however, differ from the African curve although their poles are much closer to the African curve than the European (Fig. 4). The deviation of the pole associated with the Campania platform from the African polar wandering curve is probably due to the allochthonous nature of this unit. The deviation of the Umbrian sequence pole may also be due to allochthony [22], but it is also possible that the Umbrian sequence rotated by 15°, relative to the Dolomites and Apulia, due to decoupling of the northern Apennine "autochthon" from the rest of autochthonous Italy along the Anzio–Ancona line and beneath the Po Basin.

The Umbrian sequence has also been sampled by Lowrie and Alvarez [20] who obtained very similar results; the respective circles of 95% confidence (6.5° and 5.9°) overlapping. Lowrie and Alvarez compared their results with the rather limited and suspect palaeomagnetic data from stable Europe and inferred a 43° post-Cretaceous rotation of the Italian peninsular relative to Europe. When palaeomagnetic observations are considered from all of Italy, it seems more realistic to interpret the observations in terms of autochthonous Italy having moved systematically with Africa since at least Permian times, and having therefore undergone a rotation of 50° relative to extra-Alpine Europe since Triassic times, of which 30–35° was completed by the Late Cretaceous. There have been subsequent minor rotations of the individual

TABLE 5

	Computed rotations relative to stable Europe since:			Computed rotations relative to Africa since:		
	Early Cretaceous	Middle Cretaceous	Late Cretaceous	Early Cretaceous	Middle Cretaceous	Late Cretaceous
Dolomites		+23°	+18°		+2°	+4°
Umbrian sequence		+56°	+32°		+36°	+19°
Apulia		+13°			+6°	
Campania		+32°			+13°	
Sicily [16,17]			+11.5°			− 3°
Malawi [29]	25°			−1.5°		

Clockwise rotations −ve.
Anticlockwise rotations +ve.

autochthonous blocks relative to each other, but these are essentially second-order phenomena within a pattern that is essentially African. This interpretation does not appear consistent with Soffel's observation in the southern Dolomites [6]; he suggested a 55° rotation of this area between Late Eocene and Late Miocene times. However, the difference in declination between Soffel's two groups of palaeomagnetic directions is only 45° and the more easterly group is so poorly defined ($\alpha_{95} = 24$) that the magnitude of the inferred rotation is uncertain.

The interpretation of mainland Italy as an essentially African tectonic unit is not consistent with several postulated reconstructions for the Mediterranean area, such as those based on oroclinal flexures [30] or matching bathymetric features [9, 31], but is supported by the apparent continuity of geological features from North Africa, through Italy, into the Dinarides which indicates the close proximity and essentially similar relationships of these areas to each other throughout the Mesozoic and Cenozoic, Paleomagnetic work is continuing on the allochthonous regions of the Apennines and northern Sicily to evaluate the rotations between individual allochthonous units and this may provide further evidence to allow a fuller understanding of local tectonic features and therefore of the gross behaviour of Italy. Similarly, palaeomagnetic studies are in progress using samples from Tunisia, Algeria, Greece and Israel which appear to confirm suggestions that the present bathymetry of the eastern Mediterranean must be controlled by Late Miocene and subsequent subsidence rather than by ocean-floor generation although the detailed tectonic pattern and relationships in this area have obviously been modified by recent subduction along the Hellenic and Eolian Arcs. In the western Mediterranean, modification may have occurred due to Neogene crustal accretion in the Balearic basin associated with the eastward migration of Corsica, Sardinia and Calabrian Arc and the development of the allochthonous nature of northern Sicily and the Apennines [2], but these recent modifications appear to have had little effect on the gross relationships of the area.

Acknowledgements

This represents part of a project of palaeomagnetic research in southern Europe and North Africa financed by the Natural Environment Research Council. One of the authors (J.E.T. Channell) gratefully acknowledges the support of a NERC studentship and from the proceedings of the NATO Advanced Study Publication "Implications of Continental Drift to the Earth Sciences" edited by D.H. Tarling and S.K. Runcorn. The generous help and advice given by Professor B.D'Argenio (Napoli), Professor G. Napoleoni (Firenze) and Dr. R. Catalano (Palermo) are gratefully acknowledged together with the help of many colleagues at Newcastle upon Tyne and at various Italian Institutes. We thank Professor W. Lowrie and Dr. H.C. Jenkins for their helpful advice.

References

1 W. Alvarez, Dating the rotation of Corsica–Sardinia plate. Nature 235 (1972) 58.

2 W. Alvarex, T. Cocozza and F.C. Wezel, Fragmentation of the Alpine Orogenic Belt by microplate dispersal, Nature 248 (1974) 309.

3 J.F. Dewey, W.C. Pitman, W.B.F. Ryan and J. Bonnin, Plate tectonics and evolution of the Alpine system, Geol. Soc. Am. Bull. 84 (1973) 3137.

4 M. Manzoni, Palaeomagnetic data of Middle and Upper Triassic age from the Dolomites (E. Alps, Italy), Tectonophysics 10 (1970) 411.

5 A.E.M. Nairn and M. Westphal, Possible implications of the palaeomagnetic study of Late Palaeozoic igneous rocks of NW Corsica. Palaeogeog. Palaeoclimatol. Palaeoecol. 5 (1968) 179.

6 H. Soffel, Anticlockwise rotation of Italy between Eocene and Miocene, Palaeomagnetic evidence, Earth Planet. Sci. Lett. 17 (1972) 207.

7 J.D.A. Zijderveld, G.J.A. Hazen, M. Nardin and R. Van der Voo, Shear in the Tethys and the Permian palaeomagnetism in the Southern Alps, including new results, Tectonophysics 10 (1970) 639.

8 J.D.A. Zijderveld and R. van der Voo, Palaeomagnetism in the Mediterranean Area, in: Implications of Continental Drift to the Earth Sciences, D.H. Tarling and S.K. Runcorn, eds. (Academic Press, 1973) 133.

9 A.G. Smith, Alpine deformation and the oceanic areas of the Tethys, Mediterranean and Atlantic, Geol. Soc. Am. Bull. 82 (1971) 2039.

10 W.G. Pitman III and M. Talwani, Sea-floor spreading in the North Atlantic, Geol. Soc. Am. Bull. 83 (1972) 619.

11 K.M. Storetvedt, Genesis of West Mediterranean basins, Earth. Planet. Sci. Lett. 21 (1973) 22.

12 J. de Boer, The Geology of the Vicenturian Alps with special reference to their palaeomagnetic history, Geol. Utraiectina 11 (1963) 178.

13 J. de Boer. Palaeomagnetic indications of megatectonic movement in the Tethys, J. Geophys. Res. 70 (1965) 931.

14 D. van Hilten, Geology and Permian palaeomagnetism of the Val-di-Nor area, Geol. Utraiectina 5 (1960) 95.

15 D. van Hilten, Presentation of palaeomagnetic data, polar wandering and continental drift, Am. J. Sci. 260 (1962) 401.

16 A. Schult, Palaeomagnetism of Upper Cretaceous volcanic rocks in Sicily, Earth Planet. Sci. Lett. 19 (1973) 97.

17 F. Barberi, L. Givetta, P. Gasparini, F. Innocenti, R. Scandone and L. Villari, Evolution of a section of the Africa–Europe plate boundary: palaeomagnetic and volcanological evidence from Sicily, Earth Planet. Sci. Lett. 22 (1974) 123.

18 R. Colombi, P. Giese, G. Luongo, C. Morelli, M. Rivsetti, S. Scarascia, K.G. Schutte, J. Strowald and G. Visentani, Preliminary report on the seismic refraction profile; Gargano – Salerno – Palermo – Pantelleria, Bull. Geof. Tero. Appl. (1973)

19 L. Molyneux, A complete result magnetometer for measuring the remanent magnetisation of rocks, Geophys. J. 24 (1971) 429.

20 W. Lowrie and W. Alvarez, Rotation of the Italian Peninsular, Nature 251 (1974) 285.

21 E. Abbate and M. Sagri, The eugeosynclinal sequences, in: Development of the Northern Apennines Geosyncline, G. Sestini, ed., Sediment. Geol. 4 (1970) 251.

22 V. Bortolotti, P. Passerini, M. Sagri and G. Sestini, The miogeosynclinal sequences, in: Development of the Northern Apennines Geosyncline, G. Sestini, ed., Sediment. Geol. 4 (1970) 341.

23 O. Renz, Stratigraphische und mikropaleontologische Untersuchung des Scaglia (Obere Kreide-Tertiar) im zentralen Apennin, Eclogae Geol. Helv. 29 (1936) 1.

24 O. Renz, Ricerche stratigrafiche e micropaleontologiche sulla Scaglia (Cretaceo superiore – Terziario) dell'Apennino Centrale, Mem. Descr. Carta Geol. Italia 29 (1951) 1.

25 F. Ippolito, B.D'Argenio, T. Pescatore and P. Scandone, Unita' Stratigrafico – Structurali e Schema Tectonico Dell 'Apennino Meridionale. Instituto di Geologia e Geofisica dell' Universita' Di Napoli Pubbl. N. 15 (1973).

26 B. D'Argenio, Central and Southern Italy Cretaceous bauxites, Inst. Geol. Publ. Hungarici Ann. 54/3 (1970) 221.

27 D.H. Tarling and D.T.A. Symons, A stability index of remanence in Palaeomagnetism, Geophys. J. 12 (1967) 443.

28 R. van der Voo and R.B. French, Apparent polar wandering for the Atlantic bordering continents Late Carboniferous to Eocene, Earth. Sci. Rev. 10 (1974) 99.

29 J.C. Briden, A new palaeomagnetic result from the Lower Cretaceous of East-Central Africa, Geophys. J.R. Astr. Soc. 12 (1967) 375.

30 S.W. Carey, A tectonic approach to continental drift. Symp. Continental Drift, Hobart, Tasmania (1958) 177.

31 A.G. Smith, The so-called Tethyan Ophiolites, in: Implications of Continental Drift to the Earth Sciences, D.H. Tarling and S.K. Runcorn, ed. (Academic Press, 1973) 977.